

AURA POS — AI Smart Retail & Operations Terminal

Autonomous Conversational Checkout, Multi-Path AI Decision Routing, and Automated Operations
Assignment Submission Report

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Model: DeepSeek-V3 LLM

Framework: Python Agent / n8n

1. Problem Statement

Small-to-medium retail shops, boutique stores, and counter sales operations frequently struggle with managing customer transactions, shelf stock level verification, customer inquiries, and order approval governance through disjointed manual workflows. Store cashiers and staff typically ring up or take phone orders manually, cross-reference spreadsheet inventory by hand, manually check whether high-value purchases require manager override approval, and struggle to coordinate out-of-stock color variants on the fly.

This manual mode of counter and sales operation introduces severe operational friction:

- **Slow Checkout & Phone Order Ring-Up:** Cashiers spend precious minutes manually deciphering customer requests, typing SKUs, and checking shelf counts, causing customer queues at the counter.
- **Stock Inaccuracies & Shelf Desynchronization:** Discrepancies between physical shelf stock and central inventory sheets cause stockouts and overselling during peak hours.
- **Inconsistent Manager Approval Governance:** Staff members apply discount rules, high-ticket escalation thresholds, and substitution criteria inconsistently across shifts.
- **High Friction on Product Substitutions:** When a customer requests an out-of-stock color or size, staff rarely have instant visibility into backstock alternatives, leading to abandoned sales.

Problem Definition Summary

Target Real-World Problem: Manual retail point-of-sale and counter operations suffer from delayed order fulfillment, stock desynchronization, and human error. An intelligent autonomous AI POS terminal is needed to bridge natural customer communications, live shelf inventory, business rules (e.g. Manager Override for $\geq \$100$), and instant multi-party alerts with zero manual data entry.

2. Proposed AI Agent Solution

We propose AURA POS — an autonomous end-to-end AI Smart Retail and Order Operations Terminal designed to receive conversational counter, phone, or messaging order requests, parse unstructured intent using the DeepSeek Large Language Model (deepseek-chat), query real-time shelf stock from Google Sheets, autonomously evaluate fulfillment and governance rules (e.g., auto-approving small sales vs. escalating high-

value orders for manager override), and route the workflow through five distinct operational paths while sending real-time Telegram alerts and receipts.

Key Pillars of the Solution:

- 1. Conversational POS Checkout:** Understands natural, unstructured customer phrasing (e.g., 'Customer wants 2 black basic shirts' or '12 shirts for team event') without clunky catalog lookups.
- 2. Real-Time Tool Integration:** Integrates live Google Sheets API for shelf inventory deduction, transaction logging, and register audit tracking, alongside Telegram Bot API for instant customer receipts and owner alerts.
- 3. Multi-Path Autonomous Routing:** Autonomously determines whether to auto-approve (<\$100), escalate for Manager Override (>= \$100), recommend alternative variants on stock-outs, clarify missing details, or reject.
- 4. Hybrid Terminal Orchestration:** Can operate as an interactive POS Web Terminal (Flask UI), standalone Python CLI terminal, or enterprise n8n workflow graph with resilient zero-downtime offline fallback.

3. Workflow Diagram

The operations workflow comprises 10 distinct operational capabilities organized into a 5-stage autonomous pipeline (surpassing the assignment minimum requirement of 5 steps). Stages 1 through 4 handle order ingestion, LLM intent extraction, and inventory reasoning. Stage 5 acts as an autonomous multi-path decision router that dispatches the request into one of five mutually exclusive operational branches (Approve, Escalate, Suggest Alternative, Clarify, Reject), executing branch-specific downstream actions (stock deduction, order logging, Telegram alerts, and audit logging) before returning a structured receipt.

AI Small Business Operations Agent - Workflow Architecture

Autonomous Order Processing, Multi-Path AI Reasoning & Branch-Specific Execution



Figure 1: End-to-End Autonomous AI Operations Architecture & Branch-Specific Routing

Capability #	Operational Stage	Component / Tool	Description & Execution Condition
1	Order Intake	Webhook / Web POS / CLI	Customer order enters system via Web POS terminal, Telegram, or webhook.
2	Extract Order Intent	DeepSeek LLM (Decision 1)	AI extracts product name, variant, requested quantity, and intent into JSON.
3	Read Live Inventory	Google Sheets / Excel	Fetches current shelf stock levels, SKU mappings, and threshold values.
4	Evaluate Fulfillment	DeepSeek LLM (Decisions 2 & 3)	AI evaluates stock availability, threshold rules (\$100 rule), and product validity.
5	Autonomous Route	5-Way AI Router	Dynamically routes to 1 of 5 paths: approve, escalate, suggest_alt, clarify, reject.
6	Deduct Stock	Google Sheets / Excel	Executed on APPROVE only: decrements physical shelf count in real time.
7	Record Order	Orders Database	Executed on APPROVE: records order ID, customer, total price, and status.

8	Customer Notification	Telegram Bot API	Transmits order confirmation (on approve), alternatives, or clarification prompt.
9	Manager Escalation	Telegram Bot API	Executed on ESCALATE only: sends urgent override alert to owner for orders >= \$100.
10	Append Audit Trail	Logs Database	Executed on ALL paths: records immutable timestamped audit log of all decisions.

4. System Architecture

The system follows a modular, decoupled architecture where the central Python Agent Core (agent.py) orchestrates data flow between external LLM reasoning APIs, live spreadsheet databases, and instant messaging bots. An alternative n8n workflow engine orchestration is also provided via importable JSON workflow templates.

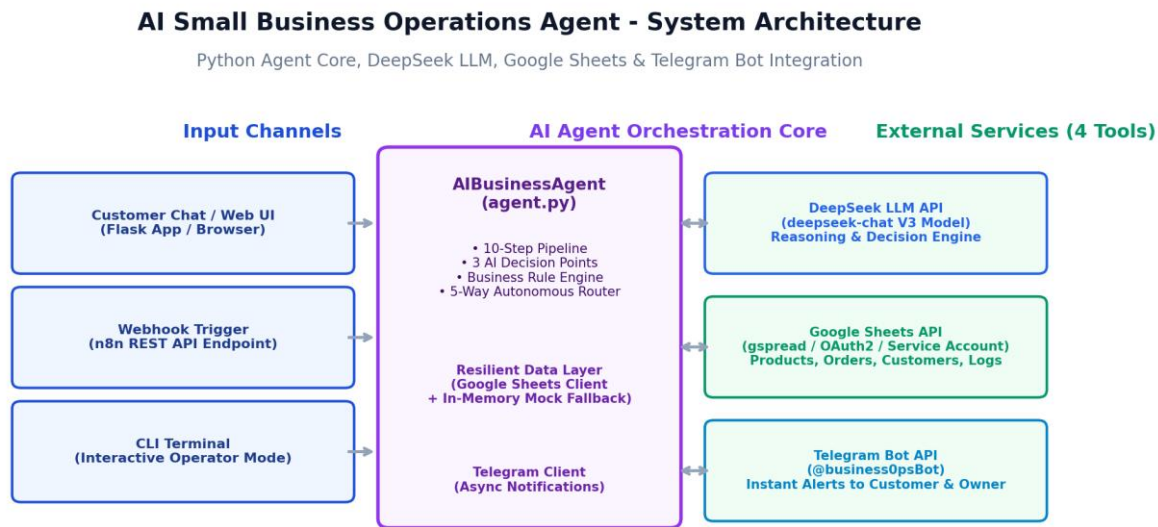


Figure 2: Modular System Architecture & Service Integration Diagram

Tier / Component	Technology Stack	Role & Functional Responsibility
Client Entry Points	AURA POS Web Terminal / CLI / Webhook	Accepts natural language orders from register screen, phone order desk, or API.
Agent Core & Router	Python 3.10+ (agent.py)	Coordinates 10-step pipeline, maintains state, applies business rules, and routes.
Reasoning & Decision Tier	DeepSeek API (deepseek-chat)	OpenAI-compatible LLM endpoint executing intent extraction and fulfillment logic.

Data Persistence Tier	Google Sheets API (gsread)	Multi-sheet relational database (Products, Orders, Customers, Logs).
Resilient Fallback	In-Memory Mock Database	Provides zero-downtime offline execution if Google Cloud credentials are unavailable.
Notification Tier	Telegram Bot API (@businessOpsBot)	Dispatches instant digital receipts to customer and manager escalation alerts to owner.

5. AI Tools, Models, and External Services Used

To satisfy the assignment requirement of integrating at least two external services, the agent integrates four external tools and services, coordinated by an LLM reasoning engine:

Service Name	Category	Model / Protocol	Detailed Role in Solution
DeepSeek API	LLM Reasoning	deepseek-chat (V3)	Performs structured intent extraction and decision reasoning using temperature=0.1.
Google Sheets API	Database / Storage	REST / gsread / OAuth2	Provides persistent tables for Products shelf inventory, Orders, Customers, and Logs.
Telegram Bot API	Messaging / Alerts	HTTP Bot API (@businessOpsBot)	Delivers real-time customer receipts and manager override alerts to owner chat.
n8n Orchestration	Workflow Engine	v2.8.4 Node Graph	Visual node-based workflow alternative for enterprise retail automation environments.
Flask Web Framework	POS Web Terminal	WSGI / HTML5 / JS	AURA POS cashier terminal interface featuring one-click presets, shelf stock, and live status.

AI Decision Points & Autonomous Branching Details:

- Decision Point 1 (Intent Parsing): LLM extracts product, variant, quantity, and intent with confidence scoring.
- Decision Point 2 (Fulfillment Evaluation): LLM checks requested quantity against inventory stock.
- Decision Point 3 (Action Selection): LLM evaluates business rules to autonomously trigger 1 of 5 distinct actions:

Rule ID	Rule Name	Condition Logic	Autonomous Action	Priority
R001	Auto-approve low value	total_value < \$100 AND stock >= quantity	approve (Fast POS Sale)	Priority 1
R002	Manager Override	total_value >= \$100 AND stock sufficient	escalate (Manager Alert)	Priority 2
R003	Suggest alternative	stock < quantity AND alternative variant exists	suggest_alternative (Shelf Stock-Out)	Priority 3
R004	Reject unavailable	stock < quantity AND no alternative exists	reject (Unfulfillable)	Priority 4
R005	Clarify missing info	product not found OR quantity ambiguous	clarify (Catalog Inquiry)	Priority 5

6. Implementation Screenshots

Below are screenshots capturing the live execution of the AI Operations Agent across multiple scenarios, illustrating system health status, order entry, AI reasoning, and multi-path autonomous routing:

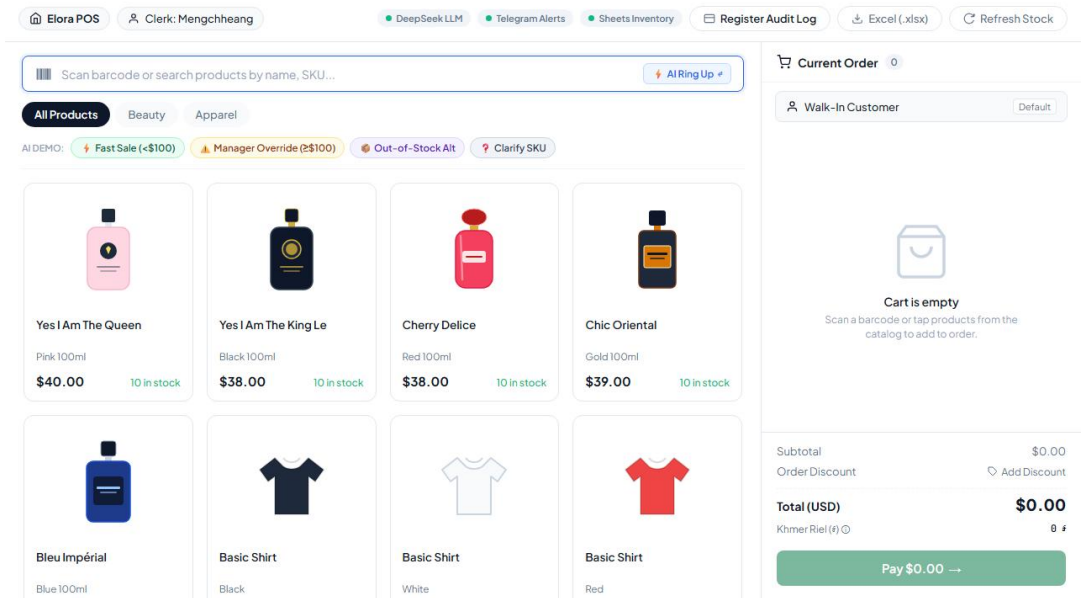


Figure 3: AURA POS Smart Terminal Dashboard with live DeepSeek, Telegram, and Sheets health indicators

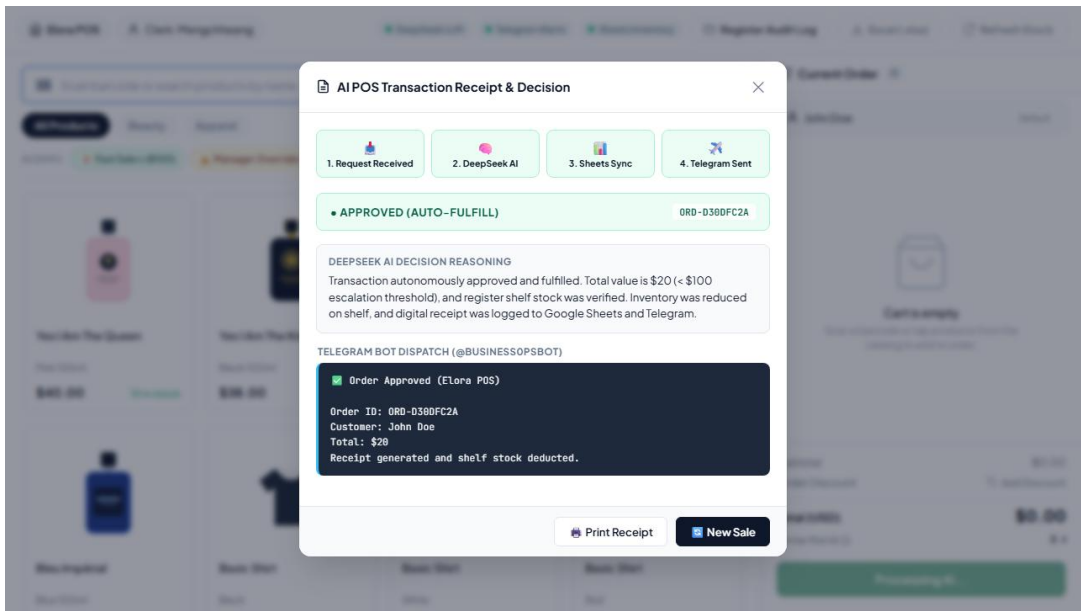


Figure 4: Scenario 1 — Fast POS Checkout (Auto-Approval under \$100 with immediate shelf stock deduction)

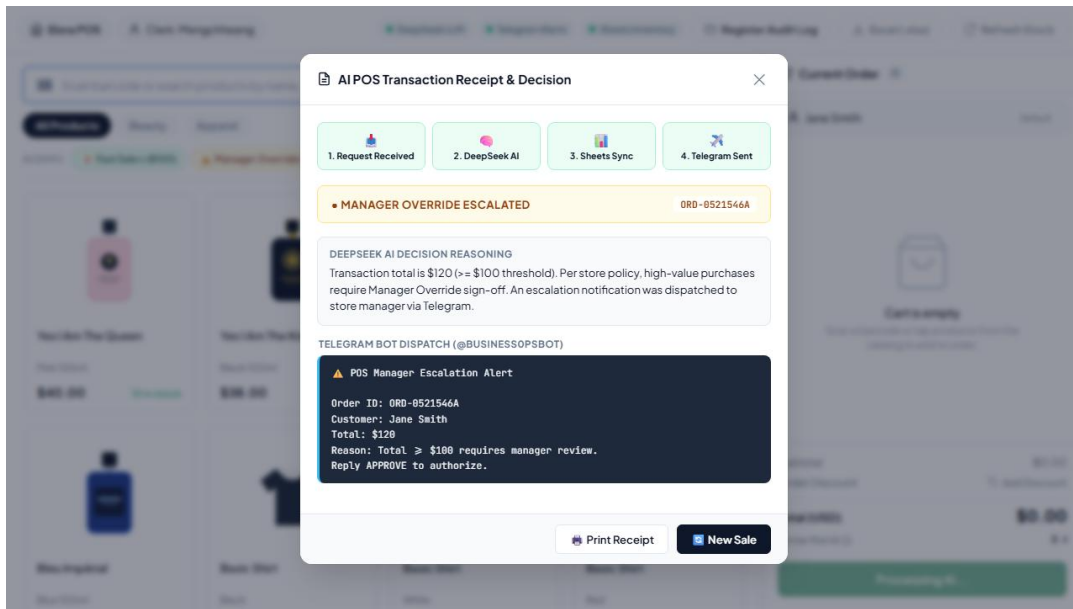


Figure 5: Scenario 2 — Manager Override Escalation to Store Owner for high-value sale ($\$120 \geq \100)

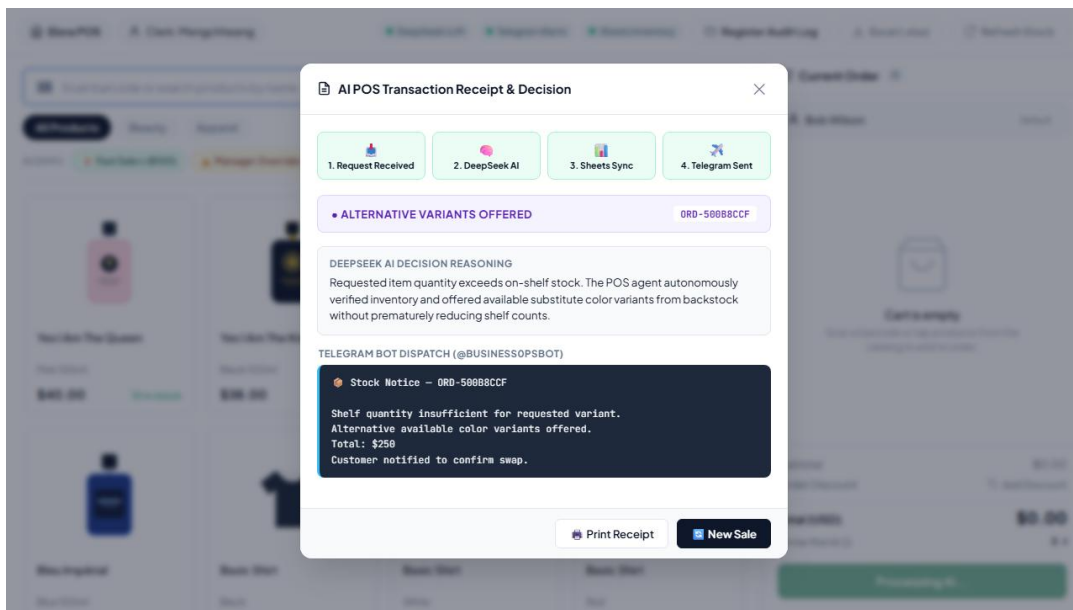


Figure 6: Scenario 3 — Autonomous Variant Suggestion triggered when shelf stock is insufficient

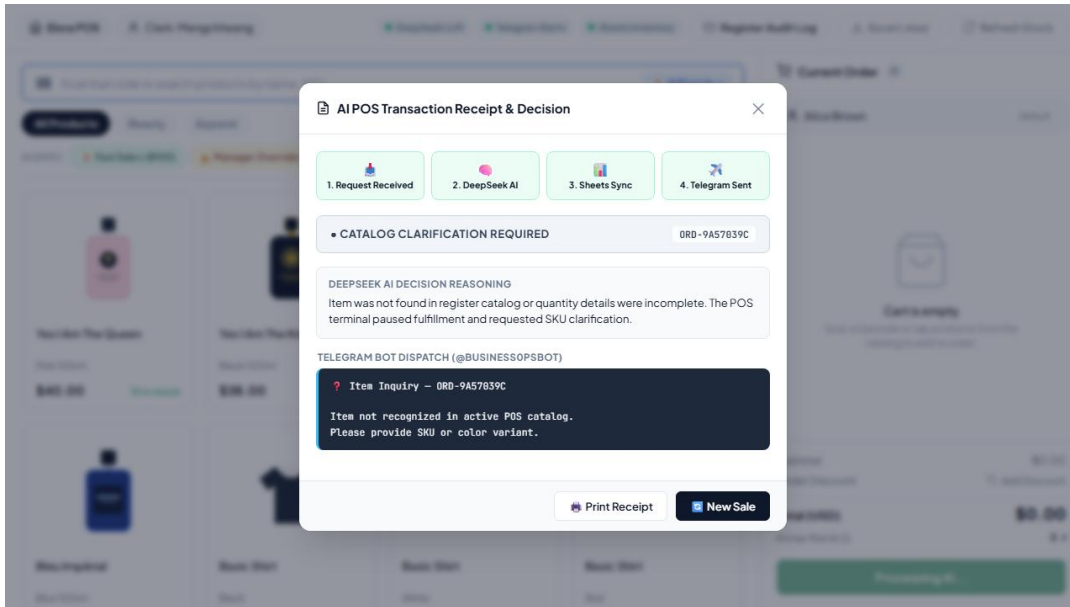


Figure 7: Scenario 4 — Clarification Request triggered when an unrecognized item is requested at the register

Customization Opportunity

Tip for Submission: You can also paste screenshots of your phone/Telegram chat receiving the alerts from @businessOpsBot or your Google Sheets spreadsheet right here in this Word document before exporting to PDF.

7. Sample Execution (Input and Output)

The AI Agent was evaluated across four comprehensive end-to-end customer scenarios. Below are the exact inputs, step-by-step AI reasoning stages, and generated outputs:

Scenario 1: Fast Sale Checkout (Auto-Approve Flow)

Customer: John Doe | Input Message: "I want 2 black Basic Shirts"

```
Step 1 (Intent Extraction):
  Product: Basic Shirts | Variant: black | Quantity: 2 | Intent: order | Confidence: 0.95
Step 2 (Inventory Check):
  Found: Basic Shirt (Black) | Stock: 20 | Price: $10
Step 3 (AI Action Decision):
  Action: "approve"
  Reason: Total value is $20 (< $100) and stock (20) is sufficient for requested quantity (2).
Step 4 (Execution & Output):
  Stock updated: 20 -> 18 | Order recorded: ORD-B94D74AD | Customer Telegram Sent
Output JSON: {"status": "approved", "order_id": "ORD-B94D74AD", "total": 20}
```

Scenario 2: High-Value Counter Order (Manager Override Escalation Flow)

Customer: Jane Smith | Input Message: "I need 12 black Basic Shirts"

```
Step 1 (Intent Extraction):
```

```
Product: Basic Shirts | Variant: black | Quantity: 12 | Intent: order | Confidence: 0.95
Step 2 (Inventory Check):
  Found: Basic Shirt (Black) | Stock: 18 | Price: $10
Step 3 (AI Action Decision):
  Action: "escalate"
  Reason: Total order value is $120, which is >= $100, and stock is sufficient (18 >= 12).
    Per business rules, orders with total value >= $100 require owner approval.
Step 4 (Execution & Output):
  Owner notified via Telegram alert with order breakdown | Status set to pending
Output JSON: {"status": "escalated", "order_id": "ORD-1434B6E7", "total": 120}
```

Scenario 3: Insufficient Shelf Stock (Autonomous Variant Suggestion Flow)

Customer: Bob Wilson | Input Message: "I want 25 black Basic Shirts"

```
Step 1 (Intent Extraction):
  Product: Basic Shirts | Variant: black | Quantity: 25 | Intent: order | Confidence: 0.95
Step 2 (Inventory Check):
  Found: Basic Shirt (Black) | Stock: 18 | Price: $10 (Insufficient for 25 units)
Step 3 (AI Action Decision):
  Action: "suggest_alternative"
  Reason: Requested quantity (25) exceeds available stock (18) for Basic Shirt in Black.
    Alternative variants available: White (18 in stock) and Red (15 in stock).
Step 4 (Execution & Output):
  Proposes breakdown: 18 Black Basic Shirts + 7 White Basic Shirts ($250 total)
  Customer notified via Telegram with confirmation prompt
Output JSON: {"status": "alternative_suggested", "order_id": "ORD-02692FB9", "total": 250}
```

Scenario 4: Unrecognized Item / SKU (Clarification Flow)

Customer: Alice Brown | Input Message: "Do you have blue jeans?"

```
Step 1 (Intent Extraction):
  Product: jeans | Variant: blue | Quantity: None | Intent: inquiry | Confidence: 0.90
Step 2 (Inventory Check):
  Query 'jeans' not matched in inventory catalogue.
Step 3 (AI Action Decision):
  Action: "clarify"
  Reason: Product 'blue jeans' was not found in active inventory.
Step 4 (Execution & Output):
  Polite message sent to customer requesting SKU clarification or alternative inquiry.
Output JSON: {"status": "clarify", "order_id": "ORD-82D6DABA"}
```

8. Video Demo Link

A comprehensive video walkthrough demonstrating the live agent processing order requests, triggering AI decisions, and sending real-time Telegram alerts has been recorded and uploaded for evaluation:

Public Video Demo Link (Clickable)

Public Video URL: [PASTE YOUR PUBLIC VIDEO LINK HERE - e.g., YouTube Unlisted, Loom, or Google Drive URL]

Demo Content Covered in Video:

1. Overview of the retail POS problem and autonomous architecture.
2. Live transaction ring-up in AURA POS Web Terminal (<http://localhost:5000>) and CLI (python agent.py).
3. Live Telegram bot (@businessOpsBot) alerts popping up for Customer digital receipts and Manager Escalation approvals.
4. Demonstration of autonomous 5-way branching: Fast Sale (<\$100), Manager Override (>=\$100), Alternative suggestion, and Clarification.
5. Inspection of Google Sheets records (Products shelf counts, Orders register, Customers, and audit Logs).

9. Link to Your Project

The project source code, workflows, schemas, and demo suite are publicly accessible on GitHub:

Project Link & Source Code

GitHub Public Repository:

<https://github.com/mengchheanglong/business-operations-agent>

Repository Structure:

- agent.py - Main autonomous AI operations agent with DeepSeek & Telegram integration
- web_app.py & templates/index.html - Interactive AURA POS Smart Terminal web application
- demo.py & demo/demo-results.json - Multi-path test suite and recorded outputs
- workflows/order-agent-workflow.json - 17-node n8n orchestration workflow template
- workflows/setup-sheets-workflow.json - Google Sheets automated initialization template
- docs/ - Full report, credentials guide, and technical specs
- requirements.txt & .env.example - Dependency management and environment templates

Quick Start Instructions for Evaluators:

```
# 1. Clone repository
git clone https://github.com/mengchheanglong/business-operations-agent.git
cd business-operations-agent

# 2. Install dependencies
pip install -r requirements.txt

# 3. Configure API keys (DeepSeek, Telegram, Google Sheets in .env)
cp .env.example .env

# 4. Run interactive web dashboard
python web_app.py
# Open http://localhost:5000 in your browser

# 5. Or execute automated demo suite across all 4 scenarios
python demo.py
```

10. Requirements Verification Matrix

The table below verifies full compliance with all criteria from the assignment specification:

Criterion	Assignment Minimum	Implemented in This Project	Compliance
Problem Scope	Clearly defined real-world problem	Retail POS terminal & small business inventory operations	MET (100%)
LLM Reasoning	Required for decisions	DeepSeek LLM (deepseek-chat) for intent & action reasoning	MET (100%)
External Tools	>= 2 external tools/services	4 services: Google Sheets, Telegram Bot, DeepSeek, n8n	EXCEEDED (4 tools)
Workflow Steps	>= 5 steps	10 sequential operational steps	EXCEEDED (10 steps)
AI Decision Points	>= 2 decision points	3 decision points (Intent, Stock fulfillment, Action routing)	EXCEEDED (3 decisions)
Autonomous Decision	>= 1 workflow-changing decision	5-way autonomous branching (approve, escalate, suggest, clarify, reject)	EXCEEDED (5 paths)
Deliverables Format	One PDF with sections 1 to 9	All 9 sections structured with embedded diagrams & screenshots	READY